

Ashwin Subramanian Murugan

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EDUCATION

Massachusetts Institute of Technology (MIT)

Expected Jun 2031

S.M. and Ph.D. in Aeronautics and Astronautics

Graduate Researcher, DNaMo Group (Prof. Hamsa Balakrishnan)

NDSEG Fellow (2026)

Indian Institute of Technology (IIT), Madras

2022 - 2026

B.Tech in Aerospace Engineering

CGPA: **9.88/10**

Department Rank: 1/75

TEST SCORES

- Secured **All India Rank 156** in JEE Advanced 2022 out of **150K+** candidates *Aug 2022*
- Secured **All India Rank 122** in JEE Mains 2022 out of **1M+** candidates *Jul 2022*
- Secured **All India Rank 239** in Kishore Vaigyanik Protsahan Yojana (KVPY SX) 2021 and awarded the KVPY Fellowship by Department of Science and Technology, Govt. of India *Jan 2021*

HONOURS AND AWARDS

- Awarded the **National Defense Science and Engineering Graduate (NDSEG) Fellowship** by the U.S. Department of Defense for graduate studies at MIT *Apr 2026*
- Awarded **Sri S Subramanian Prize** for securing the **highest CGPA** among **1500+** students in the first year of the B.Tech program at IIT Madras *Apr 2024*
- Awarded **Prof T K Varadan Prize** for achieving the best academic performance in the second year of the B.Tech Aerospace Engineering program at IIT Madras *Apr 2025*
- Awarded **Prof E G Tulapurkara Prize** for achieving the best academic performance in the third year of the B.Tech Aerospace Engineering program at IIT Madras *Apr 2026*
- Secured **Third Place** in the **IIT Madras-Jane Street Poker Bots Competition** for designing and developing a bot for a custom poker variant *Mar 2026*
- Secured **Runner-Up** in the **James Dyson Award 2024** at the National level for developing [HydroChurn](#) - a portable, self-sustaining water bottle *Sept 2024*
- Received the United Nations Millennium Fellowship 2025 to further develop [HydroChurn](#) *Aug 2025*
- Secured First Runner-Up in the **Airbus Flight Challenge 2024** at Shaastra, IIT Madras for developing a model to predict the dynamic effects of sudden braking on an airplane *Jan 2024*
- Received the Indian Academy of Sciences Summer Research Fellowship 2024 for a fully funded summer research internship at IISC Bangalore *May 2024*

PUBLICATIONS

J. Hwang, Q. Angéloz, **Ashwin Subramanian Murugan**, H. Lissek, and M. S. Sakar. “Acoustic resonators as wireless actuators in air for small-scale robots.” Under review at *Science Advances* (2026).

Ashwin Subramanian Murugan, T. M. Muruganandam. “Shock Train Stability in Scramjet Isolators under Back-Pressure Fluctuations.” Manuscript in preparation for submission to the *Journal of Fluid Mechanics* (2026).

Ashwin Subramanian Murugan, N. Srivastava, and P. A. Ramakrishna. “Numerical Investigation of Alumina-Induced Erosion and Slag Deposition in Solid Rocket Motors with Complex Geometries using a Eulerian–Eulerian Approach.” Poster presented at the *15th International High Energy Materials Conference and Exhibits (HEMCE 2026)*, Hyderabad, India, 2026.

RESEARCH AND INDUSTRY EXPERIENCE

École Polytechnique Fédérale de Lausanne (EPFL)

Project: Development of Acoustically Actuated Microflyers

Advisor: [Prof. Selman Sakar](#)

May 2025 – Jul 2025

Lausanne, Switzerland

- Designed and analyzed one of the smallest artificial flying robots, a millimetre-scale acoustically actuated microflyer using Helmholtz resonators for controlled flight as a part of the E3 Summer Research Program (2% acceptance rate)
- Fabricated microflyers with integrated propellers and resonators using a Nanoscribe two-photon polymerization system
- Performed experiments with a 40 kHz transducer array, demonstrating lift generation and controllable flight using acoustic power

Indian Institute of Technology Madras

Project: Effects of Back Pressure Fluctuation on Pseudo-Shock Waves in a Scramjet Isolator

Advisor: [Prof. T M Muruganandam](#)

Jul 2023 – Present

Chennai, India

- Investigating the effect of back pressure fluctuations on shock train stability within a scramjet isolator across varying flow conditions
- Performing high-fidelity Large Eddy Simulations (LES) using OpenSBLI to resolve unsteady shock-boundary layer interactions and pressure wave dynamics
- Working on determining Isolator design criteria to prevent unstart under fluctuating back-pressure conditions

Agnikul Cosmos

Propulsion Engineering Intern

Nov 2024 – Jan 2025

Chennai, India

Agnikul Cosmos is an IIT Madras-based aerospace startup creating 3D-printed, semi-cryogenic launch vehicles for affordable small satellite launches

- Modeled and analyzed the **LOX subcooling, pumping, and filling systems** for the booster stage testing facility using AFT Fathom, determining flow rates, pressure distributions, and valve operation parameters
- Simulated transient thermal behaviour of the LOX system using **NASA's GFSSP** to evaluate critical parameters related to the liquid nitrogen subcooling system, the liquid oxygen reservoir, and the booster stage filling system
- Analyzed **heat transfer and thermal stratification** in the booster stage propellant tanks to determine propellant requirements

Indian Institute of Science Bangalore

Project: Numerical and Experimental Study on Shock Wave Pulsations over Spiked Cylinder

Advisor: [Prof. Duvvuri Subrahmanyam](#)

May 2024 – Jul 2024

Bangalore, India

- Conducted experiments in the **Roddam Narasimha Hypersonic Wind Tunnel (RNHWT)** to investigate shock wave pulsations at **Mach 6**
- Performed hypersonic flow simulations using Ansys Fluent to analyze the dynamics of shock wave pulsations over a spiked cylinder and validated it using experimental data
- Performed a literature review to understand the driving mechanisms of shock wave pulsations

Indian Institute of Technology Madras

Project: Deposition and Erosion in Solid Rocket Motor Nozzles

Advisor: [Prof. P. A. Ramakrishna](#)

Jul 2025 – Jun 2026

Chennai, India

- Performed multiphase CFD simulations of solid rocket motor (SRM) nozzles to analyze particle-induced erosion and deposition mechanisms using empirical modeling approaches
- Developed novel wall boundary conditions for particle-resolved wall modelling in an Eulerian–Eulerian solver to study deposition and erosion
- Carried out full-scale SRM firing experiments to provide quantitative benchmarks for erosion and deposition model validation

- Developing a stereo vision-based wave reconstruction framework for estimating ocean surface dynamics from moving vessels
- Implementing motion-aware wave reconstruction and conducting virtual validation studies in Blender prior to real-world experimental deployment
- Integrating reconstructed wave fields with quiescent period prediction (QPP) models to improve prediction accuracy in marine environments

TECHNICAL PROJECTS

Course Project - Data-Driven Modelling of Aerospace Systems

Jul 2025 – Nov 2025

IIT Madras

Chennai, India

- Developed a Fourier Neural Operator-based model for the prediction of transient flows over parameterized elliptical geometries
- Designed a Reduced Order Model-based controller for morphing-airfoil actuation, enabling constant-lift performance under highly turbulent flow conditions

Course Project - 3D Compressible Curvilinear DNS Solver

Jan 2026 – May 2026

IIT Madras

Chennai, India

- Developed a 3D compressible curvilinear finite-difference DNS CFD solver for high-fidelity flow simulations
- Implemented numerical schemes for solving the compressible Navier–Stokes equations on curvilinear grids and validated the solver on canonical benchmark cases

Student Unmanned Aerial Systems Competition (SUAS) 2025

Nov 2024 – Jul 2025

Team Advisor: [Prof. Satadal Ghosh](#), IIT Madras

Chennai, India

- Part of a 15-member team developing an autonomous quadcopter UAV for SUAS 2025, focusing on autonomous flight, real-time obstacle avoidance, object detection, and precision payload delivery
- Implemented computer vision algorithms to detect, classify, and geolocate ground markers from altitudes of up to 75 feet, utilizing synthetic datasets generated in Blender
- Executed autonomous mission laps at SUAS 2025, integrating waypoint navigation, real-time object detection, and autonomous mapping of specified mission areas
- Collaborated with commercial partners to adapt and deploy this technology for domain-specific applications

AstroSim

May 2023 – Mar 2024

Physics and Astronomy Club, Centre for Innovation (CFI), IIT Madras

Chennai, India

- Developed a C++ program to numerically solve the N-body problem, investigating complex system behaviours and stellar dynamics
 - Applied advanced numerical methods and optimization techniques, including the Barnes-Hut Octree algorithm, 3-D inelastic collisions, and parallel processing, achieving an **8x** performance improvement
 - Simulated the Andromeda - Milky Way collision with over **100K+** objects, demonstrating the scalability of the system for large-scale astrophysical simulations
- Link: [Github Repository](#)

TECHNICAL SKILLS AND COURSEWORK

Programming and Software :

Python, MATLAB, C++, Ansys Fluent, L^AT_EX, SolidWorks, COMSOL, AFT Fathom, NASA's GFSSP, AutoCAD, Fusion 360

General Purpose Tools :

Microsoft Office, Blender, Canva, Adobe Illustrator

Coursework :

- Rocket Propulsion, Gas Turbines, Aerodynamics, Gas Dynamics, Vibrations, Fluid Mechanics, Thermodynamics, Flight Dynamics, Gas Dynamics Lab, Aerospace Structural Mechanics
- Computational Fluid Dynamics, Parallel Computing, Scientific Computing
- Linear Algebra, Multivariable Calculus, Complex Analysis
- Econometrics, Principles of Economics, Basic Electrical Engineering

CO-CURRICULAR ACTIVITIES

Coordinator*May 2023 – Jan 2024*

Coding and Logic Vertical, Shaastra, IIT Madras

- Coordinated a nationwide competitive programming contest for college students, managing both the preliminary and final rounds to test advanced programming and analytical skills
- Developed and curated challenging programming problems for IIT Madras' first rated CodeChef contest, attracting over **25K+** participants worldwide, and coordinated the finals held at IIT Madras

Freshman Mentor*Jul 2024 – Apr 2025*

Saathi – Student Wellness Cell, IIT Madras

- Mentored four first-year students by providing guidance on academic and personal development, supporting their transition to college, and assisting with coursework and campus resources

EXTRA-CURRICULAR ACTIVITIES

Astronomy Enthusiast

- Was a member of St. John's Public School's Astronomy Club, where I participated in hands-on STEM activities and presented weekly updates on space science
- Conducted night-sky observations using telescopes, including planetary imaging, as part of Horizon - the Physics and Astronomy Club at IIT Madras

Keyboardist and Violinist

- Completed Trinity Grade 2 in Keyboard and Grade 2 in Music Theory
- Amateur violinist specializing in Indian classical music
- Was a member of the school choir and performed multiple times in the school assembly and Annual Day